

GCP Comprehensive Review Report - CRTLPyme Project

Date: November 10, 2025

Project ID: crtlpyme-477300

Review Type: READ-ONLY Documentation

Executive Summary

The CRTLPyme project has **TWO Cloud Run services** deployed, which appears to be a **DUPLICATE DEPLOYMENT**:

1. **crtlpyme** (Main service - KEEP THIS ONE)
2. **crtlpyme-app** (Duplicate service - appears to be redundant)

Both services are running the same application but with different configurations. The main service "crtlpyme" has proper Cloud SQL connectivity, while "crtlpyme-app" does NOT have Cloud SQL connection configured.

1. CLOUD RUN SERVICES

1.1 Service: crtlpyme (MAIN SERVICE - RECOMMENDED TO KEEP)

Basic Information:

- **Service Name:** crtlpyme
- **Region:** us-central1 (Iowa)
- **URL:** <https://crtlpyme-399888129827.us-central1.run.app>
- **Status:** Active and running
- **Last Deployment:** ~54 minutes ago (from time of review)
- **Current Revision:** crtlpyme-00014-7md
- **Traffic Distribution:** 100% to latest revision

Container Configuration:

- **Image:** gcr.io/crtlpyme-477300/crtlpyme@sha256:93abc...
- **Port:** 3000
- **CPU Limit:** 2 vCPU
- **Memory Limit:** 2 GiB

Scaling Configuration:

- **Type:** Automatic
- **Min Instances:** 0
- **Max Instances:** 10
- **Concurrency:** 80 requests per instance
- **Request Timeout:** 300 seconds (5 minutes)

Authentication:

- **Public Access:** Enabled (Allow unauthenticated)
- **Ingress:** All traffic allowed

Environment Variables (7 total):

1. `DATABASE_URL` - Secret reference: `DATABASE_URL:latest`
2. `NEXTAUTH_SECRET` - Secret reference: `NEXTAUTH_SECRET:latest`
3. `NEXTAUTH_URL` - Secret reference: `NEXTAUTH_URL:latest`
4. `SENDGRID_API_KEY` - Secret reference: `SENDGRID_API_KEY:latest`
5. `SENDGRID_FROM_EMAIL` - Secret reference: `SENDGRID_FROM_EMAIL:latest`
6. `TRANSBANK_API_KEY` - Secret reference: `transbank-api-key:latest`
7. `TRANSBANK_COMMERCE_CODE` - Secret reference: `transbank-commerce-code:latest`

Cloud SQL Connections (1):

-  **Connected to:** `crtlpyme-477300:us-central1:crtlpyme-db`

Billing:

- Based on requests (pay-per-use)
- CPU allocated during startup

Current Metrics:

- **Requests:** 0 (at time of review)

1.2 Service: `crtlpyme-app` (DUPLICATE SERVICE - CANDIDATE FOR REMOVAL)

Basic Information:

- **Service Name:** `crtlpyme-app`
- **Region:** `us-central1` (Iowa)
- **URL:** `https://crtlpyme-app-399088129827.us-central1.run.app`
- **Status:** Active and running
- **Last Deployment:** ~38 minutes ago (from time of review)
- **Current Revision:** `crtlpyme-app-00001-xxx` (exact revision not captured)
- **Traffic Distribution:** 100% to latest revision

Container Configuration:

- **Image:** `gcr.io/crtlpyme-477300/crtlpyme@sha256:032d2c...`
- **Port:** 3000
- **CPU Limit:** 2 vCPU
- **Memory Limit:** 2 GiB

Scaling Configuration:

- **Type:** Automatic
- **Min Instances:** 0
- **Max Instances:** 10
- **Concurrency:** 160 requests per instance (DIFFERENT from `crtlpyme`)
- **Request Timeout:** 300 seconds (5 minutes)

Authentication:

- **Public Access:** Disabled (Requires authentication) 
- **Ingress:** All traffic allowed

Environment Variables (11 total):

1. `NODE_ENV` - Value: `production`
2. `NEXT_PUBLIC_APP_NAME` - Value: `CRTLPyme`
3. `GOOGLE_CLOUD_PROJECT_ID` - Value: `crtlpyme-477300`
4. `DATABASE_URL` - Secret reference: `DATABASE_URL:latest`
5. `NEXTAUTH_SECRET` - Secret reference: `NEXTAUTH_SECRET:latest`
6. `NEXTAUTH_URL` - Secret reference: `NEXTAUTH_URL:latest`
7. `SENDGRID_API_KEY` - Secret reference: `SENDGRID_API_KEY:latest`
8. `SENDGRID_FROM_EMAIL` - Secret reference: `SENDGRID_FROM_EMAIL:latest`
9. `TRANSBANK_API_KEY` - Secret reference: `transbank-api-key:latest`
10. `TRANSBANK_COMMERCE_CODE` - Secret reference: `transbank-commerce-code:latest`
11. `TRANSBANK_ENVIRONMENT` - Secret reference: `TRANSBANK_ENVIRONMENT:latest`

Cloud SQL Connections:

- **✗ NO Cloud SQL connection configured!** This is a critical issue.

Billing:

- Based on requests (pay-per-use)
- CPU allocated during startup

Current Metrics:

- **Requests:** 0 (at time of review)

1.3 Key Differences Between Services

Feature	crtlpyme (MAIN)	crtlpyme-app (DUPLICATE)
Authentication	Public (unauthenticated)	Requires authentication
Concurrency	80	160
Cloud SQL Connection	✓ Connected	✗ NOT Connected
Environment Variables	7 (minimal)	11 (includes <code>NODE_ENV</code> , etc.)
Container Image	Different SHA256	Different SHA256
Last Deployment	~54 min ago	~38 min ago

⚠ CRITICAL FINDING: The `crtlpyme-app` service does NOT have Cloud SQL connectivity configured, which means it cannot access the database properly. This is likely why it's not functioning correctly.

2. CLOUD SQL DATABASE

2.1 Instance Details

Basic Information:

- **Instance ID:** `crtlpyme-db`

- **Database Engine:** PostgreSQL 15.14
- **Edition:** Enterprise
- **Status:**  Running (Active)
- **Region:** us-central1-a
- **Zonal Availability:**  Single zone (no high availability)

Connection Information:

- **Instance Connection Name:** crtlpyme-477300:us-central1:crtlpyme-db
- **Public IP Address:** 136.116.45.158
- **Outgoing IP Address:** 34.28.82.6
- **Port:** 5432 (standard PostgreSQL)
- **Private IP:** Disabled
- **Public IP:** Enabled

Resource Configuration:

- **vCPUs:** 4
- **Memory:** 16 GB
- **Storage Type:** SSD
- **Storage Size:** 100 GB
- **Automatic Storage Increase:**  Enabled

Backup Configuration:

- **Automated Backups:**  Enabled
- **Backup Level:** Standard
- **Point-in-Time Recovery:**  Enabled
- **Backup Retention After Deletion:**  Enabled

Security Configuration:

- **Instance Deletion Prevention:**  Enabled
- **SSL/TLS Encryption:**  Disabled
- **Require SSL Connections Only:**  Disabled
- **Require Trusted Client Certificates:**  Disabled
- **Server Certificate Authority:** Google-managed internal CA
- **Certificate Expiration:** November 3, 2035, 2:45:59 AM

2.2 Network Security

Authorized Networks:

-  **CRITICAL SECURITY ISSUE:** Network "Acceso-Cloud-Shell" allows access from **0.0.0.0/0** (ANY IP ADDRESS)
- This means the database is publicly accessible from anywhere on the internet
- **Recommendation:** Restrict to specific IP ranges or use Cloud SQL Proxy only

Connection Methods:

- **Google Cloud Service Authorization:**  Disabled
- **App Engine Authorization:**  Disabled

2.3 Databases

The Cloud SQL instance contains **2 databases**:

1. **crtlpyme** (Application database)
 - Collation: en_US.UTF8
 - Character Set: UTF8
 - Contains 72 users (as per previous information)
 2. **postgres** (System database)
 - Collation: en_US.UTF8
 - Character Set: UTF8
 - Default PostgreSQL system database
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2.4 Database Users

Database User:

- **Username:** postgres
 - **Authentication Type:** Built-in (Integrado)
 - **Password Status:** N/A (configured)
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2.5 Instance Health Issues (3 detected)

1.  **Security - Exposure to Wide Range of Public IPs**
 - The instance allows connections from 0.0.0.0/0
 - High security risk
 - Recommendation: Restrict to specific IPs or use private IP
 2.  **Audit Not Enabled**
 - Database audit logging is not configured
 - Recommendation: Enable for compliance and security monitoring
 3.  **No User Password Policies**
 - No password complexity requirements configured
 - Recommendation: Implement password policies
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3. SECRET MANAGER

The project uses **10 secrets** stored in Google Secret Manager:

Secret Name	Purpose	Replication	Status
DATABASE_URL	PostgreSQL connection string	Automatic	Active
database-password	Database password	Automatic	Active
JWT_SECRET	JWT token signing	Automatic	Active
NEXTAUTH_SECRET	NextAuth.js secret	Automatic	Active
NEXTAUTH_URL	NextAuth.js callback URL	Automatic	Active
SENDGRID_API_KEY	SendGrid email API key	Automatic	Active
SENDGRID_FROM_EMAIL	SendGrid sender email	Automatic	Active
TRANS-BANK_ENVIRONMENT	Transbank environment (sandbox/prod)	Automatic	Active
ttransbank-api-key	Transbank payment API key	Automatic	Active
ttransbank-commerce-code	Transbank commerce code	Automatic	Active

Secret Versions:

- NEXTAUTH_URL has 4 versions (latest: version 4, created 10/11/25 6:27 PM)
- All secrets are Google-managed with automatic replication
- All secrets are in “Enabled” state

Note: Secret values are not displayed in this report for security reasons.

4. CONTAINER REGISTRY / ARTIFACT REGISTRY

4.1 Repositories

The project has **3 container repositories**:

1. cloud-run-source-deploy

- Format: Docker
- Type: Standard
- Region: us-central1 (Iowa)
- Status: Active
- Purpose: Cloud Run source deployments

2. **crtlpyme** (Main application repository)

- Format: Docker
- Type: Standard
- Region: us-central1 (Iowa)
- Status: Active
- Contains: 13 image versions
- Image sizes: 118.6 MB - 118.8 MB
- Last updated: 2 days ago

3. **gcr.io** (Legacy Container Registry)

- Format: Docker
- Type: Standard
- Region: us (multiple regions in United States)
- Status: Active

4.2 Container Images

Repository: **crtlpyme**

- **Image Name:** crtlpyme
- **Total Versions:** 13
- **Creation Date:** 2 days ago
- **Last Update:** 2 days ago
- **Size Range:** 118.6 MB - 118.8 MB

All 13 versions were created on the same day (2 days ago), suggesting multiple deployment attempts or iterations.

5. DOMAIN CONFIGURATION

Custom Domains:

- **✗ No custom domain mappings configured**
- Both services are using default Cloud Run URLs:
- crtlpyme-399888129827.us-central1.run.app
- crtlpyme-app-399088129827.us-central1.run.app

Recommendation: Configure a custom domain for production use (e.g., app.crtlpyme.cl or crtlpyme.cl)

6. IDENTIFIED ISSUES AND DUPLICATES

6.1 Duplicate Services

DUPLICATE DETECTED: Two Cloud Run services exist:

- crtlpyme (Main service with Cloud SQL connection)
- crtlpyme-app (Duplicate without Cloud SQL connection)

Analysis:

- Both services appear to run the same application

- Different container images (different SHA256 hashes)
- Different deployment times (crtlpyme is older)
- `crtlpyme-app` is missing critical Cloud SQL connection
- `crtlpyme-app` requires authentication (not publicly accessible)
- Both services have 0 requests, indicating neither is actively used

Recommendation:

- **KEEP:** `crtlpyme` service (has proper Cloud SQL connection)
 - **CONSIDER REMOVING:** `crtlpyme-app` service (missing database connection, requires auth)
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6.2 Security Issues

1. Database Publicly Accessible

- Cloud SQL allows connections from 0.0.0.0/0
- High risk of unauthorized access
- **Action Required:** Restrict to specific IPs or use private networking

2. SSL/TLS Not Enforced

- Database connections don't require SSL
- Data transmitted in plain text
- **Action Required:** Enable SSL requirement

3. No Audit Logging

- Database activities not logged
- Compliance and security monitoring gaps
- **Action Required:** Enable Cloud SQL audit logs

4. No Password Policies

- Weak passwords allowed
 - **Action Required:** Implement password complexity requirements
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6.3 Configuration Issues

1. crtlpyme-app Missing Cloud SQL Connection

- Service cannot access database
- Will fail on any database operation
- **Action Required:** Either add Cloud SQL connection or remove service

2. No High Availability

- Cloud SQL is single-zone (us-central1-a)
- No automatic failover
- **Action Required:** Consider enabling high availability for production

3. No Custom Domain

- Using default Cloud Run URLs
- Not production-ready
- **Action Required:** Configure custom domain

4. Authentication Mismatch

- `crtlpyme` allows public access

- `crtlpyme-app` requires authentication
- Inconsistent configuration
- **Action Required:** Standardize authentication approach

7. RESOURCE UTILIZATION

Current Usage (at time of review):

- **Cloud Run Requests:** 0 on both services
- **Database Connections:** Unknown (not visible in console)
- **Storage Used:** Unknown (not visible in overview)

Observation: Both services show 0 requests, suggesting:

- Services may not be actively used
- Services may have just been deployed
- There may be issues preventing access

8. COST OPTIMIZATION OPPORTUNITIES

1. Duplicate Service Removal

- Remove `crtlpyme-app` to eliminate redundant costs
- Estimated savings: ~50% of Cloud Run costs

2. Database Right-Sizing

- Current: 4 vCPUs, 16 GB RAM
- Consider monitoring actual usage and downsizing if underutilized

3. Container Image Cleanup

- 13 container images stored
- Consider implementing image retention policy
- Keep only last 3-5 versions

4. Scaling Configuration

- Min instances: 0 (good for cost)
- Max instances: 10 (may be high for small business app)
- Consider reducing max instances if not needed

9. RECOMMENDATIONS SUMMARY

Immediate Actions (High Priority)

1.  **Keep `crtlpyme` service** - It has proper Cloud SQL connection
2.  **Remove or fix `crtlpyme-app` service** - Missing database connection
3.  **Restrict database access** - Remove 0.0.0.0/0 from authorized networks
4.  **Enable SSL for database** - Encrypt data in transit
5.  **Update Transbank keys** - Use sandbox keys as requested by user

Short-term Actions (Medium Priority)

1. 🌐 **Configure custom domain** - For production readiness
2. 📊 **Enable audit logging** - For security and compliance
3. 🔒 **Implement password policies** - Strengthen security
4. 🧹 **Clean up old container images** - Reduce storage costs
5. 📝 **Document which service is production** - Clarify deployment strategy

Long-term Actions (Low Priority)

1. 🏢 **Enable high availability** - For production resilience
2. 📈 **Monitor and optimize resources** - Right-size database and Cloud Run
3. 🔒 **Implement private networking** - Use VPC for Cloud SQL
4. 🚀 **Set up CI/CD pipeline** - Automate deployments from GitHub

10. NEXT STEPS

Based on user's request to focus on **"crtlpyme"** service:

1. **Verify crtlpyme service is working**
 - Test URL: <https://crtlpyme-399888129827.us-central1.run.app>
 - Verify login functionality
 - Check database connectivity
2. **Fix authentication issues**
 - Verify NEXTAUTH_URL points to correct service
 - Test user login with existing accounts
 - Create test accounts if needed
3. **Update Transbank configuration**
 - Set sandbox API keys
 - Verify payment integration
4. **Decision on crtlpyme-app**
 - Delete if not needed
 - Or fix Cloud SQL connection if it serves a purpose
5. **Security hardening**
 - Restrict database access
 - Enable SSL
 - Enable audit logging

11. TECHNICAL SPECIFICATIONS SUMMARY

Cloud Run Service: crtlpyme

Service URL: <https://crtlpyme-399888129827.us-central1.run.app>
 Region: us-central1
 Container: gcr.io/crtlpyme-477300/crtlpyme@sha256:93abc...
 Port: 3000
 CPU: 2 vCPU
 Memory: 2 GiB
 Scaling: 0-10 instances
 Concurrency: 80
 Timeout: 300s
 Authentication: Public
 Cloud SQL:  crtlpyme-477300:us-central1:crtlpyme-db

Cloud Run Service: crtlpyme-app

Service URL: <https://crtlpyme-app-399088129827.us-central1.run.app>
 Region: us-central1
 Container: gcr.io/crtlpyme-477300/crtlpyme@sha256:032d2c...
 Port: 3000
 CPU: 2 vCPU
 Memory: 2 GiB
 Scaling: 0-10 instances
 Concurrency: 160
 Timeout: 300s
 Authentication: Required
 Cloud SQL:  NOT CONNECTED

Cloud SQL Instance: crtlpyme-db

Connection Name: crtlpyme-477300:us-central1:crtlpyme-db
 Engine: PostgreSQL 15.14
 Edition: Enterprise
 vCPUs: 4
 Memory: 16 GB
 Storage: 100 GB SSD
 Public IP: 136.116.45.158
 Port: 5432
 Databases: crtlpyme, postgres
 Zone: us-central1-a
 High Availability:  Disabled

12. CONCLUSION

The CRTLPyme project has a **duplicate deployment issue** with two Cloud Run services. The main service `crtlpyme` is properly configured with Cloud SQL connectivity, while `crtlpyme-app` is missing this critical connection and requires authentication, making it inaccessible and non-functional.

Key Findings:

-  Main service (`crtlpyme`) is properly configured
-  Duplicate service (`crtlpyme-app`) is misconfigured

-  Critical security issue: Database accessible from any IP
-  No custom domain configured
-  No high availability setup
-  Both services show 0 requests (not actively used or just deployed)

Primary Recommendation: Focus on the `crtlpyme` service, fix authentication issues, update Transbank keys to sandbox, and address security concerns. Consider removing `crtlpyme-app` service to eliminate confusion and reduce costs.

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Review Completed By: DeepAgent (Automated GCP Review)

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Project: CRTLPyme (crtlpyme-477300)